# On a construction of quadratic APN functions

Lilya Budaghyan, Claude Carlet, and Gregor Leander

***Abstract*—In a recent paper, the authors introduced a method for constructing new quadratic APN functions from known ones. Applying this method, they obtained the function $x^3 + \mathrm{tr}_n(x^9)$ which is APN over $\mathbb{F}_{2^n}$ for any positive integer $n$. The present paper is a continuation of this work. We give sufficient conditions on linear functions $L_1$ and $L_2$ from $\mathbb{F}_{2^n}$ to itself such that the function $L_1(x^3) + L_2(x^9)$ is APN over $\mathbb{F}_{2^n}$. We show that this can lead to many new cases of APN functions. In particular, we get two families of APN functions $x^3 + a^{-1}\,\mathrm{tr}_n^3(a^3x^9 + a^6x^{18})$ and $x^3 + a^{-1}\,\mathrm{tr}_n^3(a^6x^{18} + a^{12}x^{36})$ over $\mathbb{F}_{2^n}$ for any $n$ divisible by 3 and $a \in \mathbb{F}_{2^n}^*$. We prove that for $n = 9$, these families are pairwise different and differ from all previously known families of APN functions, up to the most general equivalence notion, the CCZ-equivalence. We also investigate further sufficient conditions under which the conditions on the linear functions $L_1$ and $L_2$ are satisfied.**

***Index Terms*—Almost bent, Almost perfect nonlinear, CCZ-equivalence, Nonlinearity, S-box, Vectorial Boolean function**

## I. Introduction

For any positive integer $n$, a function $F$ from the field $\mathbb{F}_{2^n}$ to itself is called *almost perfect nonlinear* (APN) if for every $a \neq 0$ and every $b$ in $\mathbb{F}_{2^n}$, the equation $F(x) + F(x+a) = b$ admits at most 2 solutions [9]. When used as S-boxes in block ciphers, APN functions oppose an optimum resistance to differential cryptanalysis (see [1]). The notion of APN function is closely connected to the notion of *almost bent* (AB) function [6], which can be described in terms of the *Walsh transform* of functions. Let $\mathrm{tr}_n(x)$ denote the trace function from $\mathbb{F}_{2^n}$ into $\mathbb{F}_2$, that is, $\mathrm{tr}_n(x) = x + x^2 + x^4 + ... + x^{2^{n-1}}$ - more generally, for any positive divisor $m$ of $n$, we shall also use the notation $\mathrm{tr}_n^m(x) = x + x^{2^m} + x^{2^{2m}} + ... + x^{2^{(n/m-1)m}}$ - then the Walsh transform of a given function $F : \mathbb{F}_{2^n} \to \mathbb{F}_{2^n}$ is the integer valued function

$$\lambda_F(a,b) = \sum_{x \in \mathbb{F}_{2^n}} (-1)^{\mathrm{tr}_n(bF(x)+ax)}, \qquad a, b \in \mathbb{F}_{2^n}.$$

Note that the Walsh transform can also be defined in the framework of functions from the vector space $\mathbb{F}_2^n$ to itself (or to another vector space) but we shall always endow here the vector space $\mathbb{F}_2^n$ with the structure of the field $\mathbb{F}_{2^n}$.

The set $\Lambda_F = \{\lambda_F(a,b) : a, b \in \mathbb{F}_{2^n}, b \neq 0\}$ is called the *Walsh spectrum* of $F$ and the set $\{|\lambda_F(a,b)| : a, b \in \mathbb{F}_{2^n}, b \neq 0\}$ the *extended Walsh spectrum* of $F$. If the Walsh spectrum of $F$ equals $\{0, \pm 2^{\frac{n+1}{2}}\}$ then the function $F$ is called AB [6]. AB functions exist for $n$ odd only and oppose an optimum resistance to linear cryptanalysis (see [8]). Besides, every AB function is APN [6], and in the $n$ odd case, any quadratic function is APN if and only if it is AB [5]. A comprehensive survey on APN and AB functions can be found in [4].

The APN and AB properties are preserved by some transformations of functions. If $F$ is an APN (respectively AB) function, $A_1, A_2$ are affine permutations and $A$ is an affine function, then the function $F' = A_1 \circ F \circ A_2 + A$ is also APN [9] (respectively AB). The functions $F$ and $F'$ are then called *extended affine equivalent* (EA-equivalent). In [5], Carlet, Charpin and Zinoviev introduced an equivalence relation of functions, more recently called *CCZ-equivalence*, which preserves APN and AB properties: two mappings $F$ and $F'$ from $\mathbb{F}_{2^n}$ to itself are called CCZ-equivalent if by some affine permutation $\mathcal{L}$ of $\mathbb{F}_{2^n}^2$ the image of the graph of $F$ is the graph of $F'$, that is, $\mathcal{L}(G_F) = G_{F'}$ where $G_F = \{(x, F(x)) \mid x \in \mathbb{F}_{2^n}\}$ and $G_{F'} = \{(x, F'(x)) \mid x \in \mathbb{F}_{2^n}\}$. Two CCZ-equivalent functions have the same extended Walsh spectrum and are both APN/AB or not. EA-equivalence is a particular case of CCZ-equivalence and any permutation is CCZ-equivalent to its inverse. In [2], it is proven that CCZ-equivalence is more general than the conjunction of the EA-equivalence of functions and the inversion of permutations.

In the recent paper [3], it is proven that the function $x^3 + \mathrm{tr}_n(x^9)$ is APN over $\mathbb{F}_{2^n}$ for any $n$, and that for almost all $n \geq 7$ it is CCZ-inequivalent to the Gold functions (and therefore, being quadratic, EA-inequivalent to power functions), to the inverse and Dobbertin functions. And (being quadratic) this function is AB for every odd $n$. The present paper generalizes the construction

$$x^3 + \mathrm{tr}_n(x^9)$$

to the form

$$F(x) = L_1(x^3) + L_2(x^9)$$

where $L_1$ and $L_2$ are linear functions from $\mathbb{F}_{2^n}$ to itself, and it studies conditions on $L_1$ and $L_2$ such that $F$ is APN. In particular, we prove that, if $n$ is even and the function $L_1(x) + L_2(x^3)$ is a permutation of $\mathbb{F}_{2^n}$, then $F$ is APN. We have an example of the fact that even slightly changing functions $L_1$ and $L_2$ can lead to new cases of APN functions: for $n = 6, 8$, and $a$ a primitive element of $\mathbb{F}_{2^n}^*$, the APN functions $x^3 + \mathrm{tr}_n(x^9)$ and $ax^3 + \mathrm{tr}_n(a^3x^9)$ are CCZ-inequivalent. Besides, we construct a few families of APN functions: let $n$ be a positive integer and $a \in \mathbb{F}_{2^n}^*$; the function

$$F_1(x) = x^3 + a^{-1}\,\mathrm{tr}_n(a^3x^9)$$

and for $n$ divisible by 3 the functions

$$\begin{aligned} F_2(x) &= x^3 + a^{-1}\,\mathrm{tr}_n^3(a^6x^{18} + a^{12}x^{36}), \\ F_3(x) &= x^3 + a^{-1}\,\mathrm{tr}_n^3(a^3x^9 + a^6x^{18}) \end{aligned}$$

Department of Informatics, University of Bergen, PB 7803, 5020 Bergen, NORWAY; e-mail: Lilya.Budaghyan@ii.uib.no

Universities of Paris 8 and Paris 13; CNRS, UMR 7539 LAGA; Address: University of Paris 8, Department of Mathematics, 2 rue de la liberté, 93526 Saint-Denis cedex 02, FRANCE; e-mail: claude.carlet@inria.fr

Department of Mathematics, Technical University of Denmark, DENMARK; e-mail: gregor.leander@rub.de

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are APN over $\mathbb{F}_{2^n}$. As mentioned above, the function $F_1$ is CCZ-inequivalent to $x^3+\mathrm{tr}_n(x^9)$ when $n=6,8$ and $a$ is a primitive element of $\mathbb{F}_{2^6}^*$, and, therefore, we can see that $F_1$ defines a new class of APN functions which includes the class of functions $x^3+\mathrm{tr}_n(x^9)$. Besides, we show that for $n=9$, the functions $F_1$, $F_2$ and $F_3$ are pairwise CCZ-inequivalent, and the functions $F_2$ and $F_3$ are CCZ-inequivalent to any function from previously known families of APN functions. Hence, $F_1$ generalizes the family of functions $x^3+\mathrm{tr}_n(x^9)$ while $F_2$ and $F_3$ define two new families of APN functions.

## II. A construction of APN functions

### A. The case $n$ even

There is a sufficient condition for a function in even number of variables of the form $x^3+L(x^9)$ (where $L$ is linear) to be APN, which explains many cases of APN functions of this form.

*Proposition 1:* Let $n$ be an even positive integer and $L$ a linear function from $\mathbb{F}_{2^n}$ to itself. If the function

$$F(x)=x+L(x^3)$$

is a permutation over $\mathbb{F}_{2^n}$ then $F(x^3)$ is APN on $\mathbb{F}_{2^n}$.

*Proof:* Since the function $F'(x)=F(x^3)$ is quadratic then it is APN if and only if, for any $a\in\mathbb{F}_{2^n}^*$, the only solutions of the linear homogeneous equation

$$F'(ax)+F'(ax+a)+F'(a)=0 \tag{1}$$

are $0,1$. We get the equation

$$a^3(x+x^2)+L(a^9(x^8+x))=0$$

and denoting $y=x+x^2$, we obtain

$$a^3y+L(a^9(y^4+y^2+y))=0$$

with $\mathrm{tr}_n(y)=0$. Hence, denoting $u=a^3y$ and

$$A_u=\left\{L\left(u^3\left(y+\frac{1}{y}+\frac{1}{y^2}\right)\right):\ y\in\mathbb{F}_{2^n}^*,\ \mathrm{tr}_n(y)=0\right\},$$

we see that if the condition

$$u\notin A_u \text{ for every } u\in\mathbb{F}_{2^n}^* \tag{2}$$

is satisfied then $F'$ is APN. Note that $A_u\subseteq\{L\left(u^3z\right):\ z\in\mathbb{F}_{2^n},\ \mathrm{tr}_n(z)=0\}$.

Since $F$ is a permutation then $F(ux)+F(ux+u)\neq 0$ for any $x$ and any nonzero $u$, that is, $u+L(u^3(x^2+x+1))\neq 0$. Thus for any nonzero $u$

$$u\notin\{L(u^3v):v\in\mathbb{F}_{2^n},\ \mathrm{tr}_n(v)=\mathrm{tr}_n(1)\}. \tag{3}$$

For $n$ even $\mathrm{tr}_n(1)=0$ and

$$u\notin A_u\subseteq\{L(u^3v):v\in\mathbb{F}_{2^n},\ \mathrm{tr}_n(v)=\mathrm{tr}_n(1)\}.$$

Hence, condition (2) is satisfied and $F'$ is indeed APN. ∎

*Remark 1:* Note that condition (2) has to be checked only for the elements $u$ in the image of $L$. For instance, in the case $L(x)=\mathrm{tr}_n(x)$, this condition has to be checked for $u=1$ only and it is obviously satisfied. This gives a simpler way of proving that $x^3+\mathrm{tr}_n(x^9)$ is APN for every $n$ (it is also easy to show that $x+\mathrm{tr}_n(x^3)$ is bijective).

Thus, to construct APN functions using Proposition 1, one has to look for permutations of the form $x+L(x^3)$ for a linear function $L$. It happens that examples of such permutations have been already found [2], in the framework of APN functions but in a different context. Note, that if $x+L(x^3)$ is a permutation then $x+a^{-1}L(a^3x^3)$ is also a permutation for every $a\neq 0$. Up to this equivalence, there are three known classes of such permutations:

- $x+\mathrm{tr}_n(x^3)$ for any even $n$.
- $x+\mathrm{tr}_n^3(x^{12}+x^6)$ for $n$ divisible by 6.
- $x+\mathrm{tr}_n^3(x^3+x^6)$ for $n$ divisible by 6.

We used Proposition 5 given below to search for more permutations of the desired type with the help of a computer. Using Proposition 5 a quite efficient backtracking search can be implemented. This approach allows to search for all such permutations up to $n=14$ within a few days. Surprisingly, it turns out that the three classes listed above already cover all the cases for $n<16$, $n$ even. We pose the following open problem.

*Open Problem 1:* Find more permutations of the form $x+L(x^3)$ not listed above or prove that the list above is already complete.

Next, we apply Proposition 1 to the second and the third classes of permutations listed above to construct APN functions (the first class gives the known APN function $x^3+\mathrm{tr}_n(x^9)$).

*Corollary 1:* Let $n$ be a positive integer divisible by 6. Then the following functions from $\mathbb{F}_{2^n}$ to itself are APN

$$x^3+\mathrm{tr}_n^3(x^{18}+x^{36}), \tag{4}$$
$$x^3+\mathrm{tr}_n^3(x^9+x^{18}). \tag{5}$$

We extend now slightly Proposition 1. Let $n$ and $L,L'$ be linear functions from $\mathbb{F}_{2^n}$ to itself. According to Remark 3 of [2], if the function $F(x)=L'(x)+L(x^3)$ is a permutation over $\mathbb{F}_{2^n}$ then $L'$ is a permutation. Hence, by Proposition 1, if $F$ is a permutation then $F(x^3)=L'(x^3)+L(x^9)$ is APN. This fact is formulated in the following proposition.

*Proposition 2:* Let $n$ be even and $L$ and $L'$ be linear functions from $\mathbb{F}_{2^n}$ to itself. If $F(x)=L'(x)+L(x^3)$ is a permutation over $\mathbb{F}_{2^n}$ then $F(x^3)$ is APN on $\mathbb{F}_{2^n}$.

For instance, for $n$ even, let $l(x)=ax+b$ with $a\in\mathbb{F}_{2^n}^*$, $b\in\mathbb{F}_{2^n}$, and let $L$ be a linear function from $\mathbb{F}_{2^n}$ to itself. If $F(x)=x+L(x^3)$ is a permutation then

$$\begin{aligned}F(l(x)) &= l(x)+L\big((l(x))^3\big)\\ &= ax+b+L(a^3x^3+a^2bx^2+ab^2x+b^3)\\ &= ax+L(a^2bx^2+ab^2x)+L(a^3x^3)+b+L(b^3)\end{aligned}$$

is a permutation too, and we can apply Proposition 2. We deduce:

*Corollary 2:* Let $n$ be even, let $a\in\mathbb{F}_{2^n}^*$, $b\in\mathbb{F}_{2^n}$, and let $L$ be a linear function from $\mathbb{F}_{2^n}$ to itself. If $x+L(x^3)$ is a permutation over $\mathbb{F}_{2^n}$ then the function

$$ax^3+L(a^3x^9+a^2bx^6+ab^2x^3)$$

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is APN on $\mathbb{F}_{2^n}$.

Surprisingly enough, this almost straightforward corollary of Proposition 1 gives new examples of APN functions for different choices of elements $a$ and $b$ as we see it below.

*Corollary 3:* Let $a \in \mathbb{F}_{2^n}^*$. For $n$ even the function

$$x^3 + a^{-1}\,\mathrm{tr}_n(a^3x^9) \tag{6}$$

and for $n$ divisible by 6 the functions

$$x^3 + a^{-1}\,\mathrm{tr}_n^3(a^6x^{18} + a^{12}x^{36}) \tag{7}$$

$$x^3 + a^{-1}\,\mathrm{tr}_n^3(a^3x^9 + a^6x^{18}) \tag{8}$$

are APN over $\mathbb{F}_{2^n}$.

Let $\alpha$ be a primitive element of $\mathbb{F}_{2^n}^*$ and $a = \alpha^{3i+j}$ for some $i$ and $0 \le j \le 2$. Then $x^3 + a^{-1}\,\mathrm{tr}_n(a^3x^9) = a^{-1}(\alpha^j y^3 + \mathrm{tr}_n(\alpha^{3j}y^9))$ where $y = \alpha^i x$. If $j = 2$ then taking $y = z^2$ we get $x^3 + a^{-1}\,\mathrm{tr}_n(a^3x^9) = a^{-1}(\alpha z^3 + \mathrm{tr}_n(\alpha^3 z^9))^2$. Hence, (6) can give at most two different cases $a = 1$ and $a = \alpha$ (since all other cases are EA-equivalent to these two as shown above). This is also true for functions (7) and (8), that is, each of them can give at most two different cases $a = 1$ and $a = \alpha$. An important observation is that for $a = 1$ and $a = \alpha$ those functions are not necessarily CCZ-equivalent. For example, consider $n = 6, 8$, then the APN function $x^3 + \mathrm{tr}_n(x^9)$ is CCZ-inequivalent to $x^3 + \alpha^{-1}\,\mathrm{tr}_n(\alpha^3x^9)$. Hence, function (6) defines a new class of APN functions which includes the class of functions $x^3 + \mathrm{tr}_n(x^9)$ from [3].

### B. The case $n$ odd

For $n$ even the function $x + \mathrm{tr}_n(x^3)$ and, when $n$ is also divisible by 3, the functions $x + \mathrm{tr}_n^3(x^3 + x^6)$ and $x + \mathrm{tr}_n^3(x^6 + x^{12})$ are permutations while they are not permutations for $n$ odd. However, as we shall see further, the corresponding functions $x^3 + \mathrm{tr}_n(x^9)$, $x^3 + \mathrm{tr}_n^3(x^9 + x^{18})$ and $x^3 + \mathrm{tr}_n^3(x^{18} + x^{36})$ are APN for both $n$ even and $n$ odd cases. Besides, Proposition 1 does not apply to the case $n$ odd. However, it can be adapted to this case. Assume first that $F(x) = x + L(x^3)$ is a permutation over the super-field $\mathbb{F}_{2^{2n}}$, then $F(x^3)$ is APN over $\mathbb{F}_{2^n}$, since we know then that, for every nonzero element $u$ of $\mathbb{F}_{2^n}$, we have $u \notin \{L(u^3v), v \in \mathbb{F}_{2^{2n}}, \mathrm{tr}_{2n}(v) = 0\}$ and that these $v$ include all the elements of $\mathbb{F}_{2^n}$ and a fortiori those of null trace $\mathrm{tr}_n$. In fact, thanks to the observations in the proof of Proposition 1, this condition can be weakened and generalized:

*Proposition 3:* Let $n$ be any positive integer and $K$ some field extension of $\mathbb{F}_{2^n}$. Let $L$ be an $\mathbb{F}_2$-linear mapping from $\mathbb{F}_{2^n}$ to $\mathbb{F}_{2^n}$ extended to an $\mathbb{F}_2$-linear mapping from $K$ to $K$. Let $E$ be a coset in $K$ of a vector space containing $L(\mathbb{F}_{2^n})$. Assume that $F(x) = x + L(x^3)$ is injective on $E$ and that the set $\{x^2 + x + 1; x \in E\}$ contains the set of elements $y$ of $\mathbb{F}_{2^n}$ such that $\mathrm{tr}_n(y) = 0$. Then $F(x^3)$ is APN over $\mathbb{F}_{2^n}$.

We can take $K = \mathbb{F}_{2^{2n}}$ and $E = w + \mathbb{F}_{2^n}$ where $w \in \mathbb{F}_4 \setminus \mathbb{F}_2$. If $F(x) = x + L(x^3)$ is injective on $E$ then $F(x^3)$ is APN on $\mathbb{F}_{2^n}$ since the set $\{x^2 + x + 1 = x + w + (x + w)^2; x \in E\}$ contains the set of elements $y$ of $\mathbb{F}_{2^n}$ such that $\mathrm{tr}_n(y) = 0$.

For instance, if $n$ is odd and divisible by 3 then the functions

$$\begin{aligned} F_1(x) &= x + \mathrm{tr}_n^3(x^6 + x^{12}), \\ F_2(x) &= x + \mathrm{tr}_n^3(x^3 + x^6) \end{aligned}$$

are injective over $w + \mathbb{F}_{2^n} \subset \mathbb{F}_{2^{2n}}$, where $w \in \mathbb{F}_4 \setminus \mathbb{F}_2$. Indeed, the function $F_2$ is injective over $w + \mathbb{F}_{2^n}$ if and only if the function $x + w + \mathrm{tr}_n^3((x + w)^3 + (x + w)^6)$ is injective over $\mathbb{F}_{2^n}$. We have

$$\begin{aligned} & x + w + \mathrm{tr}_n^3((x + w)^3 + (x + w)^6) \\ = \;& x + w + \mathrm{tr}_n^3(x^3 + x^6) + \mathrm{tr}_n^3(w^2x + w^2x^4 + w^3 + w^6) \\ = \;& x + w + \mathrm{tr}_n^3(x^3 + x^6) + \mathrm{tr}_n^3(w^2(x + x^4)) \end{aligned}$$

because $w^4 = w$ an $w^3 = w^6 = 1$. Assume that there exist $x, a \in \mathbb{F}_{2^n}$, $a \neq 0$ such that

$$\begin{aligned} & x + \mathrm{tr}_n^3(x^3 + x^6) + \mathrm{tr}_n^3(w^2(x + x^4)) \\ = \;& (x+a) + \mathrm{tr}_n^3((x+a)^3 + (x+a)^6) + \mathrm{tr}_n^3(w^2((x+a) + (x+a)^4)). \end{aligned}$$

Then we get

$$a + \mathrm{tr}_n^3(ax^2 + a^2x + a^2x^4 + a^4x^2 + a^3 + a^6) + \mathrm{tr}_n^3(w^2(a + a^4)) = 0. \tag{9}$$

Note that since $n$ is odd and $a \in \mathbb{F}_{2^n}$ then $(\mathrm{tr}_n^3(w^2(a + a^4)))^8 = \mathrm{tr}_n^3(w(a + a^4)^8) = \mathrm{tr}_n^3(w^{2^{n-3}}(a + a^4)) = \mathrm{tr}_n^3(w(a + a^4))$. Then adding (9) to its 8-th power we obtain $a + a^8 + \mathrm{tr}_n^3((w^2 + w)(a + a^4)) = 0$ which is actually $a + a^8 + \mathrm{tr}_n^3(a + a^4) = 0$ since $w^2 + w = 1$. Thus, $a + a^8 \in \mathbb{F}_8$ and $a + a^8 = (a + a^8)^8 = a^8 + a^{64}$, which gives $a = a^{64}$. Hence, $a \in \mathbb{F}_{2^6} \cap \mathbb{F}_{2^n}^*$, and since $n$ is odd then $a \in \mathbb{F}_{2^3}^*$ and

$$a + a^3 + a^6 + \mathrm{tr}_n^3(a^2x + a^2x^4 + (a + a^4)x^2) + (a + a^4)\,\mathrm{tr}_n^3(w^2) = 0.$$

Since $\mathrm{tr}_n^3(w^2) = w^i$, where $i$ equals either 1 or 2, then the previous equality gives

$$a + a^3 + a^6 + \mathrm{tr}_n^3(a^2x + a^2x^4 + (a + a^4)x^2) = (a + a^4)w^i.$$

The left side of the equality above is in $\mathbb{F}_8$, the right in $w^i \cdot \mathbb{F}_8$. Hence, $a + a^4 = 0$ and $a \in \mathbb{F}_4 \cap \mathbb{F}_{2^3}^*$, and, therefore, $a = 1$. Then

$$0 = a + a^3 + a^6 + \mathrm{tr}_n^3(a^2x + a^2x^4 + (a + a^4)x^2) = 1 + \mathrm{tr}_n^3(x + x^4).$$

The equation $1 + \mathrm{tr}_n^3(x + x^4) = 0$ has no solutions in $\mathbb{F}_{2^n}$. Indeed, if

$$\mathrm{tr}_n^3(x + x^4) = 1 \tag{10}$$

then

$$\mathrm{tr}_n^3(x + x^2) = (\mathrm{tr}_n^3(x + x^4))^2 = 1, \tag{11}$$

and adding (10) and (11)

$$0 = \mathrm{tr}_n^3(x + x^4 + x + x^2) = \mathrm{tr}_n^3(x^2 + x^4) = (\mathrm{tr}_n^3(x + x^2))^2 = 1.$$

A contradiction.

Thus, the function $F_2$ is indeed injective over $w + \mathbb{F}_{2^n} \subset \mathbb{F}_{2^{2n}}$, and the proof for the case of the function $F_1$ is similar. Therefore, by Proposition 3 the functions $F_1(x^3)$ and $F_2(x^3)$ are APN over $\mathbb{F}_{2^n}$ for $n$ odd as well. Now we can formulate the following result.

*Corollary 4:* Let $n$ be a positive integer and $a \in \mathbb{F}_{2^n}^*$. Then function (6) is APN over $\mathbb{F}_{2^n}$, and, if $n$ is divisible by 3, functions (7) and (8) are APN over $\mathbb{F}_{2^n}$.

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Unlike to the case $n$ even, for $n$ odd different choices of $a \in \mathbb{F}_{2^n}^*$ for function (6) (respectively, functions (7) and (8)) give functions EA-equivalent to the case $a = 1$. Indeed, $\gcd(n, 3) = 1$ for $n$ odd and, for example, $ax^3 + \mathrm{tr}_n(a^3x^9) = y^3 + \mathrm{tr}_n(y^9)$ with $y = a^{\frac{1}{3}}x$.

We checked with a computer (by testing the isomorphy of the associated binary codes, see [5]) that, for $n = 9$, functions (6), (7) and (8) are pairwise CCZ-inequivalent, and, that (7) and (8) are CCZ-inequivalent to any function from previously known families of APN functions. Hence, functions (7) and (8) define two new families of APN functions.

## III. Sufficient conditions for permutations $x + L(x^3)$

To construct new APN functions thanks to Proposition 1, we need to find more linear mappings $L$ such that $x+L(x^3)$ is a permutation. There is a simple sufficient condition for that.

*Proposition 4:* Let $n$ be a positive integer and $L$ be a linear function from $\mathbb{F}_{2^n}$ to itself. If for every $u \in \mathbb{F}_{2^n}$ such that $L(u) \neq 0$, the condition

$$\mathrm{tr}_n\left(\frac{u}{(L(u))^3}\right) = \begin{cases} 0 \text{ if } n \text{ is odd} \\ 1 \text{ if } n \text{ is even} \end{cases}$$

is satisfied, then $F(x) = x + L(x^3)$ is a permutation.

*Proof:* Let $x, y \in \mathbb{F}_{2^n}$ be any elements such that

$$y = x + L(x^3). \tag{12}$$

Then, denoting $L^i(x) = (L(x))^i$, $i = 2, 3$, we have

$$x^3 = (y + L(x^3))^3 = y^3 + y^2L(x^3) + yL^2(x^3) + L^3(x^3).$$

This means that there exists $z \in \mathbb{F}_{2^m}$ such that:

$$z = (y + L(z))^3 = y^3 + y^2L(z) + yL^2(z) + L^3(z). \tag{13}$$

Then, $F$ is a permutation if and only if, for every $y$, Equation (13) has at most one solution $z$. This is clear if $n$ is odd since the function $x \to x^3$ being then bijective, Relation (13) is equivalent to (12) with $z = x^3$. If $n$ is even, then if for some $y$, there exist several $z$ satisfying (13), then for each of them, $x = y + L(z)$ has $z$ for cube and is therefore a solution of equation $y = x + L(x^3)$, which means that $F$ is not injective; conversely, if for some $y$ there exist several $x$ satisfying (12), these elements cannot have same cube because of (12) and there are several $z$ satisfying (13).

A sufficient condition for equation (13) to have at most one solution for each $y$ is that, for every $u \neq 0$, there does not exist $y$ and $z$ such that

$$u = y^2L(u) + yL^2(u) + L(z)L^2(u) + L^2(z)L(u) + L^3(u),$$

since if $z$ and $z + u$ are solutions of (13), then this equality is satisfied. Note that $L(u)$ cannot be null since $u \neq 0$. We deduce that if, for every $u \neq 0$ such that $L(u) \neq 0$, there do not exist $y$ and $z$ such that

$$\frac{u}{L^3(u)} = \left(\frac{y}{L(u)}\right)^2 + \frac{y}{L(u)} + \left(\frac{L(z)}{L(u)}\right)^2 + \frac{L(z)}{L(u)} + 1$$

then $F$ is a permutation. Hence a sufficient condition for $F$ to be a permutation is $\mathrm{tr}_n\left(\frac{u}{L^3(u)}\right) = 1$, if $n$ is even and $\mathrm{tr}_n\left(\frac{u}{L^3(u)}\right) = 0$, if $n$ is odd. ■

We shall give now a necessary and sufficient condition for $x + L(x^3)$ being bijective which is in particular helpful for a computer search. This proposition is actually a special case of Proposition 6 of [2] for which we give a slightly different proof. We first need to recall some definitions and facts. The Hamming weight $wt(f)$ of a Boolean function $f : \mathbb{F}_{2^n} \to \mathbb{F}_2$ is the size of its support $\{x \in \mathbb{F}_{2^n} : f(x) \neq 0\}$. A Boolean function $f$ is called balanced if $wt(f) = 2^{n-1}$. A Boolean function $f$ is called bent if

$$\lambda_f(a) := \sum_{x \in \mathbb{F}_{2^n}} (-1)^{f(x)+\mathrm{tr}_n(ax)} = \pm 2^{\frac{n}{2}}$$

for any $a \in \mathbb{F}_{2^n}$. Bent functions are never balanced since $f$ is balanced if and only if $\lambda_f(0) = 0$.

*Proposition 5:* Let $n$ be an even integer and $L$ be a linear function from $\mathbb{F}_{2^n}$ to itself. The function $x + L(x^3)$ is a permutation of $\mathbb{F}_{2^n}$ if and only if, for every $b \in \mathbb{F}_{2^n}^*$, such that $L^*(b) \neq 0$ there exists an element $\gamma \in \mathbb{F}_{2^n}$ such that $L^*(b) = \gamma^3$ and $\mathrm{tr}_n^2(\gamma^{-1}b) \neq 0$, were $L^*$ denotes the adjoint linear mapping of $L$.

*Proof:* The mapping $F(x) = L(x^3)+x$ is a permutation if and only if, for every $b \neq 0$, the Boolean function $\mathrm{tr}_n(bF(x))$ is balanced. We calculate

$$\begin{aligned} \lambda_F(0, b) &:= \sum_x (-1)^{\mathrm{tr}_n(b(L(x^3)+x))} \\ &= \sum_x (-1)^{\mathrm{tr}_n(L^*(b)x^3+bx)} \\ &= \sum_x (-1)^{\mathrm{tr}_n(\frac{L^*(b)}{b^3}x^3+x)} = \lambda_{x^3}\Big(1, \frac{L^*(b)}{b^3}\Big). \end{aligned}$$

Now we know that, for every $\beta \in \mathbb{F}_{2^n}$, the mapping $\mathrm{tr}_n(\beta x^3)$ is bent if and only if $\beta$ is a non-cube (see [7]), and this implies that $\lambda_{x^3}(1, \beta)$ is not zero in this case. Thus a necessary condition for $L(x^3) + x$ to be a permutation is that $\frac{L^*(b)}{b^3}$ is a cube for every $b$. This is equivalent to $L^*(b)$ being a cube for every $b$. Denote by $\gamma$ any third root of $L^*(b)$, i.e. $\gamma^3 = L^*(b)$. The next step is to actually compute the value $\lambda_{x^3}(1, \beta)$ in this case. To simplify notation we denote $L^*(b)/b^3$ by $\beta$. Then

$$\begin{aligned} \lambda_{x^3}(1, \beta)^2 &= \sum_{x,y} (-1)^{\mathrm{tr}_n(\beta x^3+x+\beta y^3+y)} \\ &= \sum_y (-1)^{\mathrm{tr}_n(\beta y^3+y)} \sum_x (-1)^{\mathrm{tr}_n(\beta(y^2x+yx^2))} \\ &\quad \text{(by replacing } y \text{ by } x+y) \\ &= \sum_y (-1)^{\mathrm{tr}_n(\beta y^3+y)} \sum_x (-1)^{\mathrm{tr}_n((\beta^2y^4+\beta y)x^2)}. \end{aligned}$$

Now the sum over x is 0 if $\beta^2y^4 + \beta y \neq 0$ and $2^n$ otherwise. We denote $B = \gamma^{-1}b \cdot \mathbb{F}_4$. Since $(\gamma b^{-1})^3 = \beta$ then $\beta^2y^4 +$

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$\beta y = 0$ if and only if $y \in B$. Then, using $\mathrm{tr}_n(1) = 0$, we get

$$\begin{aligned}\lambda_{x^3}(1,\beta)^2 &= 2^n \sum_{y\in B}(-1)^{\mathrm{tr}_n(\beta y^3+y)} = 2^n \sum_{y\in B}(-1)^{\mathrm{tr}_n(y)} \\ &= 2^n \sum_{y\in \mathbb{F}_4}(-1)^{\mathrm{tr}_n(\gamma^{-1}by)} \\ &= 2^n \sum_{y\in \mathbb{F}_4}(-1)^{\mathrm{tr}_2(\mathrm{tr}_n^2(\gamma^{-1}b)y)} \\ &= \begin{cases} 2^{n+2} & \text{if} \quad \mathrm{tr}_n^2(\gamma^{-1}b) = 0 \\ 0 & \text{else} \end{cases}.\end{aligned}$$

This completes the proof. ∎

A very similar characterization holds in the case $n$ odd.

*Proposition 6:* Let $n$ be an odd integer and $L$ be a linear function from $\mathbb{F}_{2^n}$ to itself. The function $x + L(x^3)$ is a permutation of $\mathbb{F}_{2^n}$ if and only if for all $b \in \mathbb{F}_{2^n}$ either $L^*(b) = 0$ or $\mathrm{tr}_n(\gamma^{-1}b) = 0$, were $L^*(b) = \gamma^3$ and $L^*$ denotes the adjoint linear mapping of $L$.

*Proof:* Again, the mapping $F(x) = L(x^3) + x$ is a permutation if and only if, for every $b \neq 0$, the Boolean function $\mathrm{tr}_n(bF(x))$ is balanced. A similar calculation as in the proof of Proposition 5 yields

$$\begin{aligned}\lambda_F(0,b) &= \sum_x (-1)^{\mathrm{tr}_n(b(L(x^3)+x))} \\ &= \sum_x (-1)^{\mathrm{tr}_n((L^*(b)^{1/3}x)^3+bx)} \\ &= \sum_x (-1)^{\mathrm{tr}_n\left(x^3+\frac{b}{L^*(b)^{1/3}}x\right)} \\ &= \lambda_{x^3}\left(\frac{b}{L^*(b)^{1/3}},1\right).\end{aligned}$$

Now in the case $n$ odd $\lambda_{x^3}\left(a,1\right) = 0$ if and only if $\mathrm{tr}_n(a) = 0$ (see for example [4]) and thus $F$ is a permutation if and only if $\mathrm{tr}_n\left(\frac{b}{L^*(b)^{1/3}}\right) = 0$ for all $b$ such that $L^*(b) \neq 0$. This completes the proof. ∎

A little more can be said in the case where all coefficients of $L$ are in the subfield $\mathbb{F}_{2^{n/2}}$, as it is the case for all the examples we know. For this we first state a technical lemma.

*Lemma 1:* Let $n$ be an even integer and $L$ a linear function from $\mathbb{F}_{2^n}$ to itself with $L(x) = \sum_{i=0}^{n-1}\lambda_i x^{2^i}$ where all $\lambda_i \in \mathbb{F}_{2^{n/2}}$. Furthermore denote by $L'$ the restriction of $L$ to $\mathbb{F}_{2^{n/2}}$. It holds that $L'^*(x) = L^*(x)$ for all $x \in \mathbb{F}_{2^{n/2}}$, i.e. the restriction of the adjoint operator of $L$ to $\mathbb{F}_{2^{n/2}}$ is the adjoint operator of the restriction of $L$ to $\mathbb{F}_{2^{n/2}}$.

*Proof:* First note that when all $\lambda_i$ are in the subfield, so are the coefficients of $L^*$. Now let $x, y \in \mathbb{F}_{2^{n/2}}$ be arbitrary and choose $z \in \mathbb{F}_{2^n}$ such that $y = z^{2^{n/2}} + z$. We compute

$$\begin{aligned}\mathrm{tr}_{n/2}(xL'(y)) &= \mathrm{tr}_{n/2}(xL(y)) = \mathrm{tr}_{n/2}(xL(z+z^{2^{n/2}})) \\ &= \mathrm{tr}_{n/2}(xL(z) + (xL(z))^{2^{n/2}}) \\ &= \mathrm{tr}_n(xL(z)) = \mathrm{tr}_n(L^*(x)z) \\ &= \mathrm{tr}_{n/2}(L^*(x)(z+z^{2^{n/2}})) \\ &= \mathrm{tr}_{n/2}(L^*(x)y)\end{aligned}$$

As $L'^*$ is the unique linear mapping fulfilling $\mathrm{tr}_{n/2}(xL'(y)) = \mathrm{tr}_{n/2}(L'^*(x)y)$ the lemma follows. ∎

*Proposition 7:* Let $n$ be an even integer and $L$ be a linear function from $\mathbb{F}_{2^n}$ to itself such that the function $x + L(x^3)$ is a permutation. If all coefficients of $L$ are in $\mathbb{F}_{2^{n/2}}$ then $L$ fulfills $L(\mathbb{F}_{2^{n/2}}) = \{0\}$ and $L^2 = 0$.

*Proof:* In this case $F(\mathbb{F}_{2^{n/2}}) \subseteq \mathbb{F}_{2^{n/2}}$ and therefore $F$ is also bijective on this subfield. We first prove that $L^*(b) = 0$ for all $b \in \mathbb{F}_{2^{n/2}}$ and therefore that $L(b) = 0$ for all $b \in \mathbb{F}_{2^{n/2}}$. We consider the two cases $n/2$ even and odd separately. If $n/2$ is even then applying Proposition 5 on $x + L(x^3)$ viewed as a permutation on $\mathbb{F}_{2^n}$ gives $L^*(b) = 0$ or $L^*(b) = \gamma^3$ and $\mathrm{tr}_n^2(\gamma^{-1}b) \neq 0$. But for $b \in \mathbb{F}_{2^{n/2}}$ the second condition is impossible as when $\gamma^3$ is in $\mathbb{F}_{2^{n/2}}$ so is $\gamma$ and therefore $\mathrm{tr}_n^2(\gamma^{-1}b) = \mathrm{tr}_{n/2}^2(\gamma^{-1}b\,\mathrm{tr}_n^{n/2}(1)) = 0$. Thus we see that $L^*(b) = 0$ for all $b \in \mathbb{F}_{2^{n/2}}$. In the case $n/2$ odd we apply Proposition 6 on $x + L(x^3)$ viewed as a permutation on $\mathbb{F}_{2^{n/2}}$ (and for $b \in \mathbb{F}_{2^{n/2}}$) and conclude that either $L^*(b) = 0$ or $L^*(b) = \gamma^3$ where $\mathrm{tr}_{n/2}(\gamma^{-1}b) = 0$. On the other hand applying Proposition 5 on $x + L(x^3)$ viewed as a permutation on $\mathbb{F}_{2^n}$ and Lemma 1 yields in the case where $L^*(b) \neq 0$ that $\mathrm{tr}_n^2(\gamma^{-1}b) \neq 0$, a contradiction as for an element $a \in \mathbb{F}_{2^{n/2}}$ we have $\mathrm{tr}_{n/2}(a) = \mathrm{tr}_n^2(a)$. Thus for all even $n$ it holds that $L^*(b) = 0$ for all $b \in \mathbb{F}_{2^{n/2}}$. This implies that $L(b) = 0$ for all $b \in \mathbb{F}_{2^{n/2}}$. Now, in both cases ($n$ even and $n$ odd), we choose $x \in \mathbb{F}_{2^n}$ arbitrarily and compute $L(x) + L(x)^{2^{n/2}} = L(x) + L(x^{2^{n/2}}) = L(x + x^{2^{n/2}}) = 0$ which is equivalent to $L(x) \in \mathbb{F}_{2^{n/2}}$. Hence, $L(L(x)) = 0$ for any $x \in \mathbb{F}_{2^n}$. ∎

Finally, we recall a characterization of permutation of the form $F(x) = L(x^3) + L'(x)$ given in [2].

*Proposition 8 (Prop. 5 of [2]):* Let $L$ and $L'$ be linear functions from $\mathbb{F}_{2^n}$ to itself. The mapping $F(x) = L(x^3) + L'(x)$ is a permutation over $\mathbb{F}_{2^n}$ if and only if, for every $u \in \mathbb{F}_{2^n}^*$ and every $v$ such that $\mathrm{tr}_n(v) = \mathrm{tr}_n(1)$, the condition $L(u^3v) \neq L'(u)$ holds.

## REFERENCES


[1] E. Biham and A. Shamir. Differential Cryptanalysis of DES-like Cryptosystems. *Journal of Cryptology*, vol. 4, No.1, pp. 3-72, 1991.

[2] L. Budaghyan, C. Carlet, A. Pott. New Classes of Almost Bent and Almost Perfect Nonlinear Functions. *IEEE Trans. Inform. Theory*, vol. 52, no. 3, pp. 1141-1152, March 2006.

[3] L. Budaghyan, C. Carlet, G. Leander. Constructing new APN functions from known ones. *Finite Fields and Their Applications*, v. 15, issue 2, pp. 150-159, April 2009.

[4] C. Carlet. Vectorial Boolean Functions for Cryptography. Chapter of the monography *Boolean Methods and Models*, Y. Crama and P. Hammer eds, Cambridge University Press, to appear soon. Preliminary version available at http://www-rocq.inria.fr/codes/Claude.Carlet/pubs.html

[5] C. Carlet, P. Charpin and V. Zinoviev. Codes, bent functions and permutations suitable for DES-like cryptosystems. *Designs, Codes and Cryptography*, 15(2), pp. 125-156, 1998.

[6] F. Chabaud and S. Vaudenay. Links between differential and linear cryptanalysis, *Advances in Cryptology -EUROCRYPT'94, LNCS*, Springer-Verlag, New York, 950, pp. 356-365, 1995.

[7] G. Leander. Monomial bent functions. *IEEE Transactions on Information Theory*, vol. 52, no. 2, pp. 738-743, 2006.

[8] M. Matsui. Linear cryptanalysis method for DES cipher. *Advances in Cryptology-EUROCRYPT'93, LNCS*, Springer-Verlag, pp. 386-397, 1994.

[9] K. Nyberg. Differentially uniform mappings for cryptography, *Advances in Cryptography, EUROCRYPT'93, LNCS*, Springer-Verlag, New York, 765, pp. 55-64, 1994.


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